Research status & verification

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Rank-seven multiplicity-five continuation and current verification scope

Read these results and data with caution.
AI-generated research · independent verification pending. The results, proofs, code and data have not yet been independently verified by human experts or formalized in Lean. Use them with caution; exact checks establish only their stated computational scope.

The current classification record

The rank-seven multiplicity-five continuation addresses all 5,805 supplied exact-multiplicity-five rings: 5,804 general exclusions, 1 realization and 0 unresolved cases. The classification of the supplied census is complete through multiplicity 5. The new realization R7_5_5678 is Rep(AGL(1,7)). The two earlier rank-six categorification problems also remain open. The historical 38-case continuation has 36 decisions; its inventory is separate from this extension.

Rank	Bound	Rings	Excluded	Models	Open
4	128	889,530	889,515	15	0
5	20	40,999	40,982	17	0
6	9	25,421	25,362	57	2
7	5	10,006	9,976	30	0
8	2	973	896	77	0

Ranks two, three and four are classified without a multiplicity restriction. Exactly fifteen rank-four Grothendieck rings occur, and every rank-four categorification admits a spherical structure. The table records the separate finite censuses.

The bound is the largest multiplicity included in the displayed census, not necessarily a complete classification range. Rank six is complete through multiplicity three; its exact bands 5, 6, 7 and 9 are also complete, while bands 4 and 8 each contain one open case.

Rank eight is complete through multiplicity two: 48 Grothendieck rings at exact multiplicity one and 29 at exact multiplicity two. The latter band contains 848 exclusions and 29 unitary models.

Rank-six models comprise 56 unitary constructions and one spherical construction that is not pseudo-unitary. Completeness means coverage of the stated census by the project's recorded arguments; it does not assert independent human or formal verification.

The rank-seven M5 tensor identities, fusion axioms and preservation of the earlier tensor set passed independent computational checks. Completeness of the exhaustive enumeration relies on the supplied producer completion record; the optimized enumerator was not independently rerun.

Public atlas | GitHub repository

MATHEMATICAL FRONTIER

Two open cases

Each question concerns general complex fusion categorification. The reductions below are necessary conditions with explicit limits.

R6_4_155 / Odd cyclic Clifford constraints

Every odd generalized pivotal rotation must factor through a cyclic spin character. All 1,144 tested blocks pass. The all-order path-Clifford divisibility condition for the source family holds exactly at even parameter values, including the parameter four of this case.

The unique rank-26 modular candidate and rank-22 based-module witness remain necessary data. No compatible associators, coherent module category or general obstruction is established.

R6_8_1660 / Complete indicators and central phase masses

In the positive spherical representative, saturated modular representations determine all indicator sequences and every central phase mass. The eight spin objects form four dual pairs, all with restriction 2X+2gX+sigma+3T.

Exact phase Gram constraints leave 640 labelled restriction-data profiles in three branches of 68, 286 and 286. These profiles are neither categories nor complete modular fusion data. They do not construct compatible associators or exclude all categorifications.

New structural criteria

Rank-seven multiplicity-five continuation

The new continuation combines quadratic Casimir compression, complete signed sphericalization, Galois reduction to positive dimensions, and exact simultaneous first and second center-twist moments. A separate integral case is excluded by a complete cyclic-module Morita-dual calculation and descent to dimension 148. The current manuscript gives the categorical reductions and the finite domains checked by the programs.

The center-trace certificates impose no arbitrary cutoff on root orders. The conductor-sixty extension pairs relative-trace values under modular Galois symmetry; it does not assume equality of individual paired twists. Structural reductions and unsuccessful model searches alone do not change a case to excluded.

A general equality criterion applies at any prime p congruent to one modulo four: a twist-fixed quadratic-torus vector with saturated compression eigenvalue plus or minus 1/sqrt(p) generates a representation of level p, and its nontrivial twist support lies in the corresponding Legendre class of primitive pth roots. The manuscript proves this at every prime-power level and applies p=5 to the quadratic pair.

A second uniform criterion treats the odd conductor-eight character. Its partial projections give exact mass ratios for two-primary twist orders one, two, four and eight. Duality and Galois actions fixing the complete source field then yield two finite alternatives for active twist orders. Both criteria are proved at all congruence levels and supply necessary conditions for the complete finite-certificate arguments.

R7_5_5039 is now excluded in the general complex scope by coupled fourth and eighth rotations and an integral fusion-norm inequality. An all-order relative-phase inventory covers arbitrary twist orders. Two conductor-eight branches have exact modular contradictions; the complementary branches reduce exhaustively to 48 integer trees with 96 exact separating leaves. The eighth indicators force the finite packet domain, while the final separators use fourth-rotation and fusion-integrality constraints. This is a complete source-bound argument, distinct from rejecting an individual necessary-data witness.

R7_5_3895 is also excluded in the general complex scope. Complete signed and Galois reductions, both low induction frames, every second-indicator alternative and both character labellings are covered. The final coupled-rotation systems have 240 fourth-rotation choices: 225 have exact rational contradictions, and the remaining fifteen have complete integer chains with 2,126 exact leaves. A finite integral-order calculation checks all low words without a degree cutoff and proves the necessary norm-slack parity. The complete source-bound verifier passes in normal and optimized Python with SciPy unavailable, including a repeated run in one process.

Rank eight: an inner-square obstruction

For R8_2_24, a hypothetical complex categorification has a pseudo-unitary Galois conjugate with a rank-four neutral component. Saturated modular compression forces its second indicators. Conjugation by an odd invertible would fix every neutral simple label and square to conjugation by the neutral involution.

A simultaneous pivotal chart and the short weights of associator coefficients show that such a square root would yield infinitely many inequivalent tensor functors between fixed fusion categories. Tensor-functor rigidity excludes this, giving a general complex obstruction. The source and finite-weight checks support the argument within their stated scope; its categorical reasoning remains subject to independent mathematical verification.

> **Status discipline.** A restricted nonexistence result applies only under its stated assumptions. An exact computation of conditional equations does not discharge the missing categorical hypothesis.

EXACT COMPUTATIONAL EVIDENCE

What was checked

Rank seven: multiplicity-five extension

The extension is kept separate from all frozen earlier proof stages. The preserved M4 ledger is restored for historical replay; current status partitions and reader downloads include the M5 records.

Exact normal and optimized aggregate replays passed for the current ledger.

Run supplements/rank7/software/verify_m5.py for the extension and consult its normal and optimized receipts under supplements/rank7/output/m5/. These checks certify the implemented input identities, criterion certificates and preserved historical decisions; they do not independently prove categorical coherence or exhaustive-enumeration completeness.

Ranks four and five

The unrestricted rank-four proof gives FPdim < 1000 and multiplicity at most 31 without assuming pivotality. Its exact finite classification checks 1,692 rings, 1,677 exclusions and 15 models. Every rank-four complex fusion category admits a spherical structure; the sharp bounds are FPdim ≤ $24+12\sqrt{3}$ and multiplicity ≤ 6. The theoretical arguments and exact computations have distinct verification obligations.

Both complete rank-four replays passed: 889,530 rings, 15 models and 889,515 exclusions. Each regenerated 6,470,006 signed orbits; an independent full unsigned recount also passed.

The reconstructed rank-five package passed all 40,999 pinned classes, 17 model matches and 40,982 general exclusions in normal and optimized Python. The two reconstructed ledgers agree byte-for-byte. The unavailable source ledger was not recovered; the reconstructed ledger has its own recorded digest.

Ranks seven and eight

The complete base-certificate loops passed in normal Python. Separate source-bound transition replays and current-ledger checks passed in normal and optimized Python with exact parent restoration. These checks account for the current partitions displayed in this note.

The rank-eight base replay checks all 973 records, including 75 signed proof trees, four rank-seven subring proof transfers and six appendix tensor transfers. The R8_2_24 rigidity package separately checks its canonical tensor, neutral fusion rules and finite associator-weight claims; the categorical proof has separate AI audits.

Rank-six necessary conditions

The odd cyclic Clifford checker passes all 1,144 tested blocks and its invalid-input controls. Independent exact indicator and phase-mass checks cover both conjugate choices for the second residual; the complete profile allocation leaves 640 labelled possibilities. Normal and optimized replays pass within this necessary-condition scope.

Rank six: a combined verification campaign

The complete initial process checked all 25,362 exclusion certificates: 25,355 passed and seven failed. Each of those seven then passed the complete unchanged checker under SymPy's Python backend. Exact matrix comparison attributes the failures to equation ordering or orientation and positional separating-vector conventions; no proof inequality or certificate was relaxed.

Both post-loop checks skipped by the initial failure gate passed separately. The model registry gives 57 matches, and the multiplicity-one partition gives 39 = 18 + 21. A hash-bound aggregate validates complete certificate coverage and 299 source/input bindings.

This is a combined campaign, not a successful single full run or an all-Python full-census pass. The initial process exited with failure, and that status is preserved in the evidence. Its loaded backend/version was not measured at startup; the global python-flint installation changed from 0.8.0 to 0.9.0 during the run, so worker and child processes may have loaded different versions.

The complete-replay orchestrator has passed identity and rejection-control checks but has not itself been executed end to end. Its existence must not be reported as an additional full-census success.

READING AND REPRODUCING THE RECORD

Evidence and its limits

Source identity

A checksum binds a statement or calculation to specified bytes. It cannot establish that their mathematics is correct.

Exact replay

A program verifies its implemented finite equations, enumerations or tensor comparisons. Its mathematical assumptions remain part of the proof obligation.

Categorical argument

A claimed construction or obstruction must establish the hypotheses connecting the finite calculation to a fusion category. Source and arithmetic checks support a categorical proof only within their stated scope.

Independent review

Separate AI review passes are not independent human verification. No human verification or Lean formalization is claimed here.

Replay from the repository root

```
python3 tools/restore_data.py
python3 tools/verify_current_publication.py
python3 -O tools/verify_current_publication.py
```

Add --historical to run the complete source-bound certificate chain on isolated inputs. Checkers using assert require normal Python; an optimized invocation alone removes those assertions. The current-publication verifier also checks the identity and consistency of the published documents and their website copies.

The rank-specific full-census commands, dependencies, receipts and limitations are documented in the verification guide. The aggregate release checks are separate from the mathematical certificate replays.

Current sources

Research status
Verification guide
Public research status page

The manuscript statements, individual ledger entries and their source-bound evidence should be read together. This note summarizes the current scope; it is not an independent proof certificate.